# Demo

Doing linear regression

# Interpreting Linear Regression

Let's say this is our linear regression model for testing the association between systolic blood pressure and diastolic blood pressure:

```
Call:
lm(formula = bp_sys ~ bp_dia, data = hrs_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-66.649 -10.297  -1.396   9.561  76.099

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 38.71302    2.03510   19.02   <2e-16 ***
bp_dia       1.21980    0.02595   47.00   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 15.39 on 2353 degrees of freedom
Multiple R-squared:  0.4842,  Adjusted R-squared:  0.484
F-statistic:  2209 on 1 and 2353 DF,  p-value: < 2.2e-16
```

Outcome (Dependent Variable)

Predictor (Independent Variable)

```
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```

## For each 1 mmHg increase in diastolic blood pressure, systolic blood pressure is expected to be 1.22 mmHg higher, on average.

```
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```

**Null hypothesis:** The coefficient is zero (no linear association).
**Alternative hypothesis:** The coefficient is not zero (a linear association exists).

# Demo

Multivariable regression

# Knowledge Check

**True or False:** According to the model, females have higher systolic blood pressure than males, after accounting for diastolic blood pressure and age.

```
Call:
lm(formula = bp_sys ~ bp_dia + age_2014 + female, data = hrs_clean)

Residuals:
    Min      1Q  Median      3Q     Max
-72.256  -9.745  -1.361   8.584  75.172

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7.89278    5.04964  -1.563    0.118
bp_dia       1.24778    0.02516  49.602  < 2e-16 ***
age_2014     0.61138    0.05843  10.464  < 2e-16 ***
female      -3.37507    0.64733  -5.214 2.01e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 14.86 on 2351 degrees of freedom
Multiple R-squared:  0.5195,  Adjusted R-squared:  0.5189
F-statistic: 847.3 on 3 and 2351 DF,  p-value: < 2.2e-16
```

**a) True**

**b) False**

# Knowledge Check

**True or False:** According to the model, females have higher systolic blood pressure than males, after accounting for diastolic blood pressure and age.

```
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lm(formula = bp_sys ~ bp_dia + age_2014 + female, data = hrs_clean)

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(Intercept) -7.89278    5.04964  -1.563    0.118
bp_dia       1.24778    0.02516  49.602  < 2e-16 ***
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F-statistic: 847.3 on 3 and 2351 DF,  p-value: < 2.2e-16
```

**a) True**

**b) False**

Females have significantly lower systolic blood pressure on average (coefficient is negative, $p<0.001$)




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